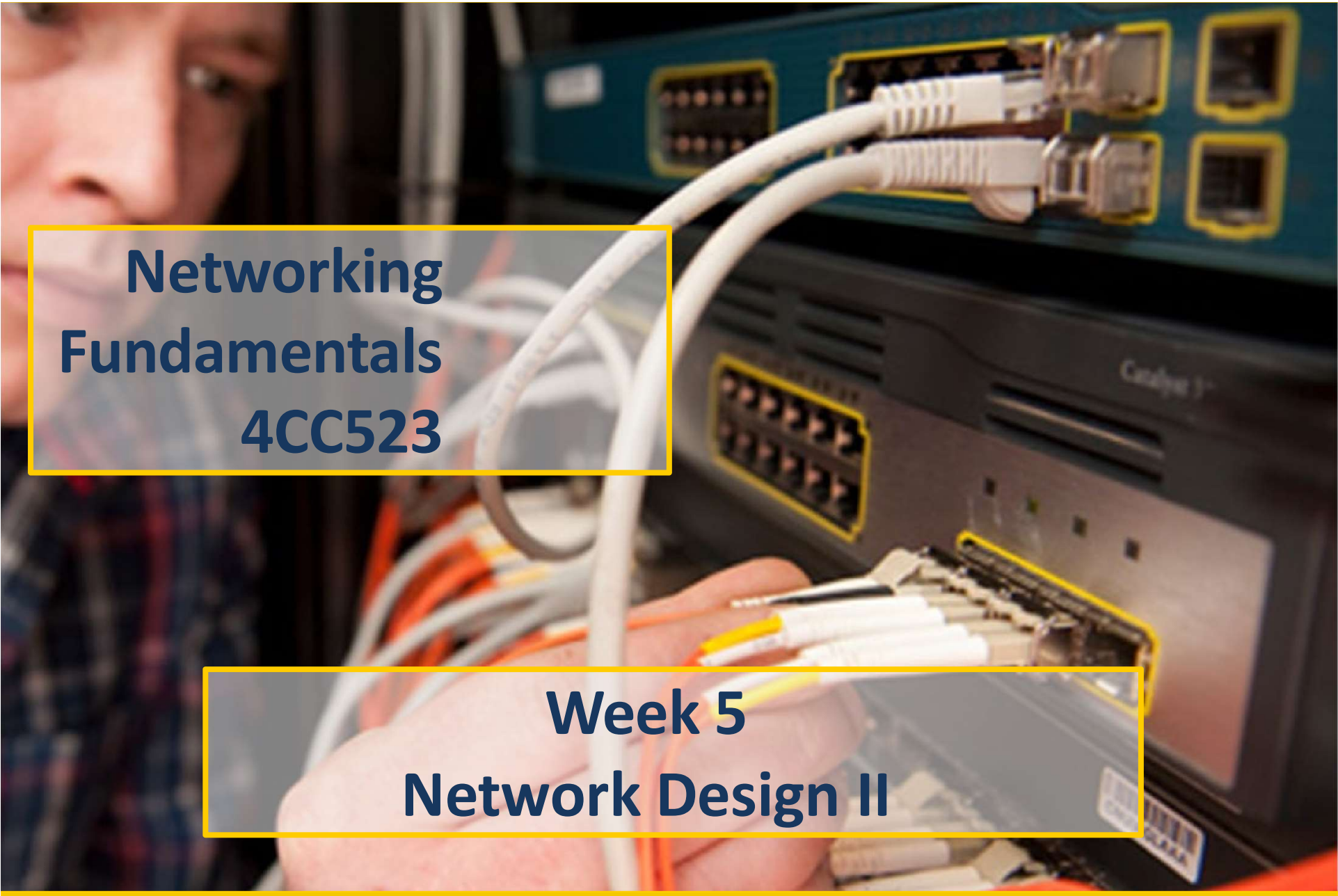


A close-up photograph of a person's face, partially visible on the left, looking towards the camera. In the background, there is a network switch or router with several white Ethernet cables plugged into its ports. A hand is visible in the foreground, holding a bundle of white and orange cables. The text "Networking Fundamentals 4CC523" is overlaid on the image in a yellow box.

Networking Fundamentals 4CC523

Week 5 Network Design II

Individual Health & Safety Responsibilities

To help prevent the spread of the COVID-19 virus we ask you to carry out the following, to help keep everyone safe & aid with contact tracing.

- If you have any COVID-19 symptoms leave the class immediately if possible & return to your accommodation. Contact the NHS on 111 & then contact student wellbeing.
- Wash your hands or sanitise them upon entering/exiting the area.
- Sign in/register via the Tap In system or alternative system in place.
- Identify a seating area & log the desk number on 'Who Sat Where?' on the Udo app. For people who don't have access to the app inform your lecturer to use the alternative system.
- Always maintain social distancing.
- When the class finishes, please use the cleaning materials provided to clean down the area you have used & then wash or sanitise your hands.
- We welcome the wearing of face coverings in teaching areas where possible.

Thanks for your help and continued support.

IMPORTANT INFO!!

–BE AWARE – BIG BROTHER IS WATCHING AND LISTENING...

- Actually listening but not watching
- Inform me if you object to recordings being made (and remind me every session!!)

–TAP, TAP – only you and do stay for the lecture

Week 5: Objectives

- Logical Design of a Network
- Physical Design
- Equipment enclosures
- Trunking and Trays
- Flood wiring / Building wiring
- Expanding the LAN
- Resilience and Redundancy
- Budgets

Theoretical network

- Most networks are actually networks of networks, the Internet is a network of networks.
- Easiest way to think of a network is to think of a cloud
- From week 1 we considered clouds and how they could be linked together
- Also considered how devices inside the cloud connect and communicate

Theoretical network

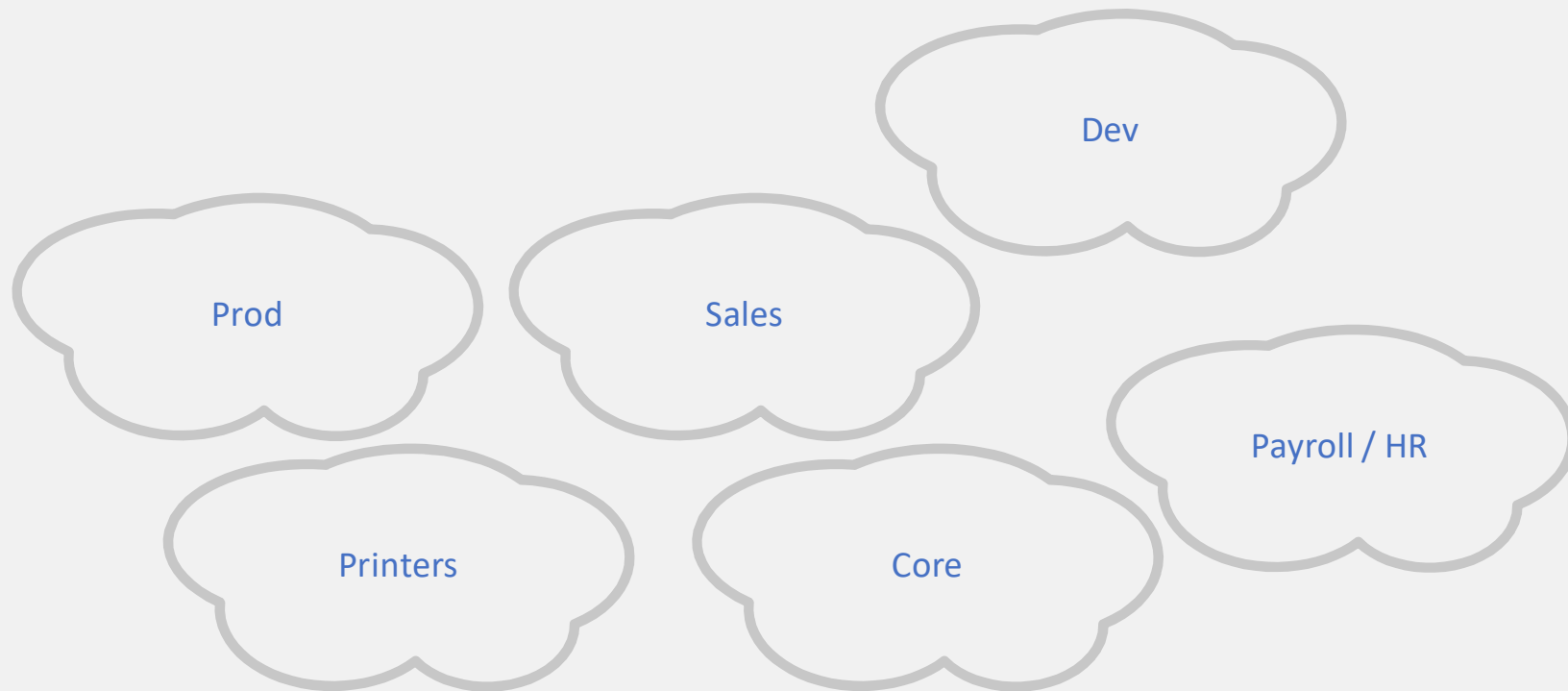
- We connect the devices using a very clever (Layer2 switch) to form a cloud, we connect the clouds together using amazingly clever (Layer3 device[?]), we protect our network from the outside world with a firewall and then connect our network to the outside world with another device, typically a router (required if we are going to change protocols e.g. DSL broadband to Ethernet)
- What could be easier???

Network Spec

- Imagine we are designing a network for a small engineering company
 - 4 departments: Sales, Development, Production, Payroll/HR
 - 18 users in each department
 - One printer on each floor
 - All user devices, servers are Microsoft based
 - Private IP addressing internally (we will get back to this)
 - Network services (DHCP, DNS, Authentication, File Storage) provided in core network
 - DSL broadband Internet connection
 - Building has 7 rooms over 4 floors, (1)reception and (2)production on ground, (3) production on 1st, (4) development and (5) sales on 2nd, (6) payroll and HR on 3rd. There is a secure storage room (7) accessible through payroll and HR on 3rd
 - Note Reception is part of Sales

Departments / LAN

- 4 departments: Sales, Development, Production, Payroll/HR plus printers and core

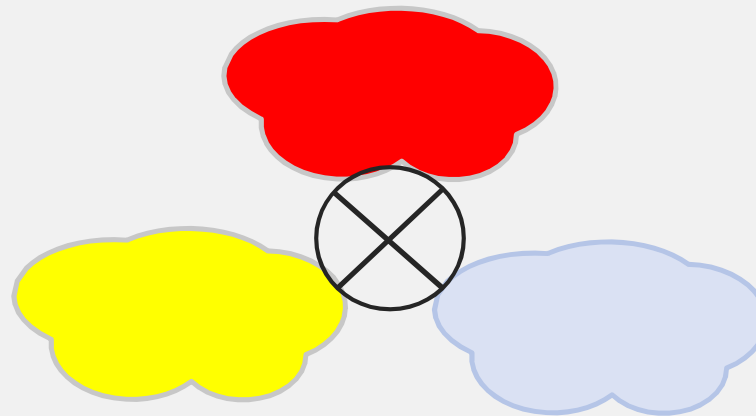


Example of one dept – Sales / HR

- 18 user devices
- Uplink connection to core network
- Redundant uplink connection to core
- Needs Layer2 switch with min 20 ports
- Could use Cisco 2960 with 24 FastEthernet plus 2 GigabitEthernet for uplinks
 - Is 2960 still available?
 - Alternative Cisco 1000?
 - Price difference between 24 FE and 24 GE port switches?

Joining clouds

- For each cloud to talk to each cloud they need to be joined by an Amazingly Clever (Layer 3 device) from week 1



Joining clouds

- Layer 3 device could be a Router or a Layer 3 switch
- Each LAN cloud will connect to (include or contain) this device (these are the gateway ports)
- Device must have 6 ports plus the outbound Internet connections (and scope for future development)
- The LAN (Layer 2) switch uplinks are GigabitEthernet, the ports on this L3 device must also be Gigabit to make use of this high bandwidth
- Try to find a Router with at least 6 Gigabit Ethernet ports at a reasonable price (!)

Joining clouds

- For resilience we used two uplinks from each Access (LAN Cloud) switch, would be wasted if these went to the same Distribution Layer device
- We need two of this device – this will be very expensive if we used a router
- Layer3 switch instead, will link all LAN clouds, need more than 6 ports, use 2 x 12 port – Cisco 3660 is FastEthernet – updated version for GE ports? Cisco 250? You choose!

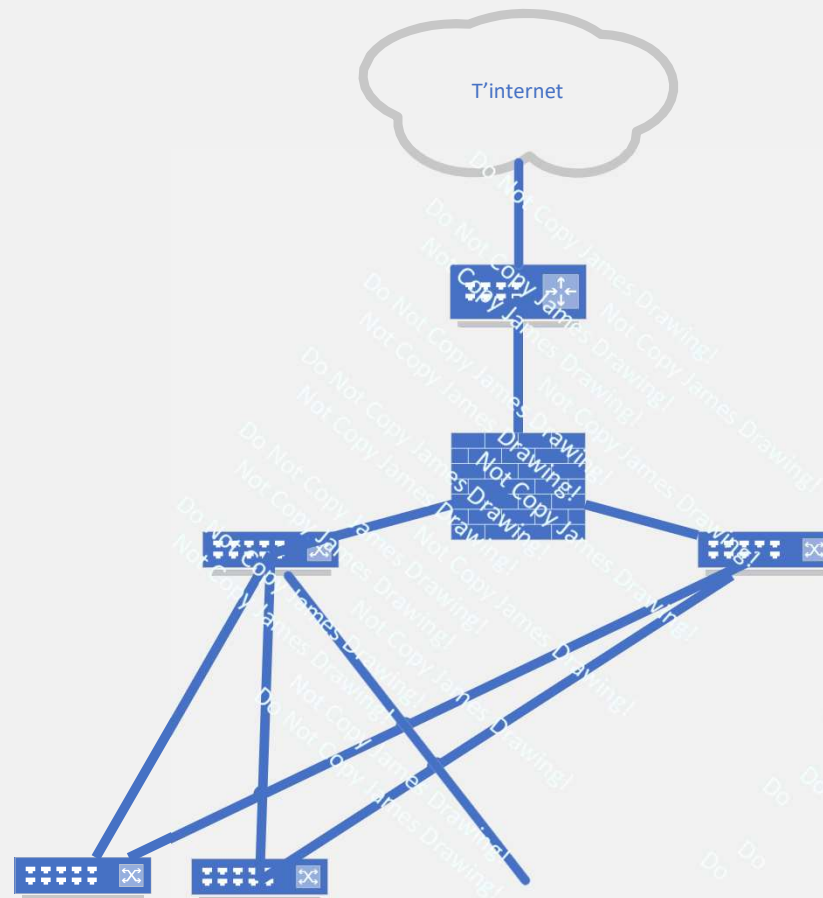
Link to outside world

- You need protection from the outside world (where all the bad people are [where the ones who do not work for you are])
- Protection is needed to prevent unauthorised access
 - Unauthorised access means that your personal or private data can be accessed
 - Also means that your network (the one with your name and your contact details and your personally identifiable information) could be used by someone (pretending to be you) to attack, abuse or defraud someone else
 - Also means that someone could set up loans in your name using your house, car, business, grandmother, they get the upfront ready, you get the bill forever
- You need protection!

Protection

- Protection is usually in the form of a Firewall
- Firewall sits between your network and the Internet (where the bad people are) preventing unauthorised connections inbound to your network
- Firewalls are most often LAN connected inside (trusted network) and outside (untrusted network)
- Might need a further router to connect the outside Ethernet based network to the ADSL based Internet connection
- ISR (Integrated Services Routers) may also contain firewall features (replace / enhance firewalls)
- You may want to add firewalls between departments e.g. to prevent your trusted employees from stealing your current developments or adding a pay rise to their salary

So far (logical) design



So far (logical) design

- Servers are all connected to “core”
- Each dept has its own switch (own cloud, own LAN)
 - Needs IP addressing information adding to diagram
- Printers have their own switch
- All switches have capacity for current use and expansion

Physical Network

- For physical protection against damage, locate equipment (servers, routers, switches) in physical containment (data cabinets)
- For protection against unauthorised physical access, locate equipment in secure area with limited access
 - Ideally in an area where any access would not only be noticed but would be challenged
- In our design that area on the third floor with the accountants would be ideal

Physical Network

- Will all equipment fit in a single cabinet? Would that be a good idea? How much 240v power would be needed in the cabinet? How much heat would there be?
- We could put a cabinet on each floor with one or more switches to suit but that means having two switches for sales / reception and two for production (costs are increasing, complexity too)
- If we keep all the switches together, it just means that we need to terminate the user connections in one place (somehow...)

Physical Network Connectivity Methods

- Start with the layer 2 (Access) to layer 3 (Distribution) connections, all inside cabinet.
 - Short distance
 - Highly reliable
 - High bandwidth and high utilisation (data compression point)
 - Resilience to interference from Radio Frequency and Electro Magnetics could be beneficial
- Could be twisted pair copper or Fibre ? Check the specs, which is best?
 - Multimode or Monomode?
 - Affects choice of L2 interface, affects choice of L3 switch

Physical Network Connectivity Methods

- Servers in Core network – very similar consideration.
 - Distance, reliability, utilisation, RFI, EMI resilience
- Twisted pair Copper or Fibre ?
 - Multimode or Monomode?
 - Affects choice of L2 **switch**, could become specialist device
 - Might need to consider another layer of redundancy in design (more ports needed on distribution switch?)
 - Will need to negotiate with server providers or at least state clearly in design!

User connections

- The connection at the PC, Printer, IP phone, WAP is very likely to be twisted pair copper.
- WAP an IP phone very often use “Power over Ethernet”
 - very convenient, accessible wherever the network connection is located
 - Does not require separate 240v socket outlet nearby
 - Does not require an Electrical contractor for installation or annual safety inspection
- Choice of connection medium is probably limited!

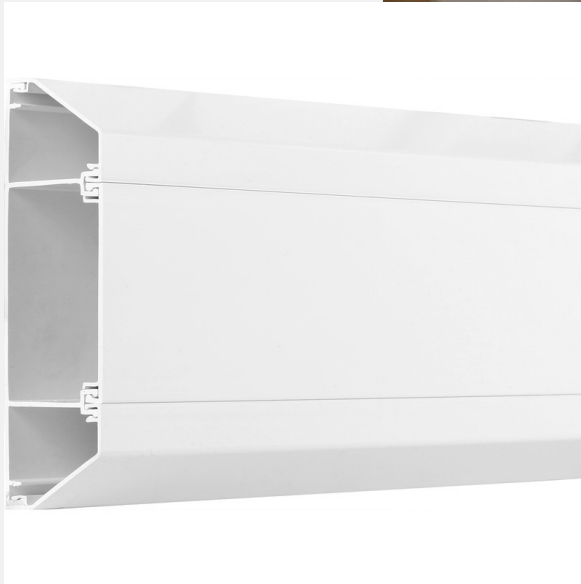
Flood Wiring

- Most new buildings will incorporate “services” (Gas, water, electricity) during the build but these will only be to a metering device
- The building owner will specify where services need to be provided for end use (kitchens, sinks, toilets, power outlets, *Service Provider Internet access points*)
- Very commonly in commercial / industrial buildings, the Electrical contractors and Network Installation contractors (if different) will agree on who will be providing Trunking and Traywork and where this will be installed

Cable Containment - Trunking

- Two or three compartment trunking is preformed hollow “box section” containment with removeable lids or covers
- Available in various heights and depths
- It is usually installed horizontally on the internal faces of all walls in a room
- Installing slightly above desk height means that it does not interfere with desk positions but is easily accessible
- Above desk height is usually also below window height, meaning that the Trunking can be installed in single “runs”

Cable Containment - Trunking



Cable Containment - Trunking

- You cannot mount Trunking across doorways!
- Trunking can also be used at skirting level (against the floor)
- Rooms can be linked “back to back” by cutting large diameter holes inside the trunking
- Trunking does not usually have a high capacity for cables, you cannot add unlimited amounts of data points
- At some point there will need to be a vertical trunking section where cables will pass into a ceiling void or to a floor above or below

Cable Containment – Cable Tray

- Where higher capacity for cabling is required, trunking will normally lead to a space between the visible (false) ceiling and the actual ceiling (very often concrete in commercial buildings). This is known as the ceiling void.
- Cable tray is open flat metal surfaces that cables are tied to using ty-rap type fixings
- Cable tray can also be “cable basket”
- There may be firebreak walls above fire doors but, in general, cable trays will span entire lengths of buildings
 - You can see them in the ground floor of the stem building or very often just by looking up when you are in Sainsburys or any number of shops in the Eagle Centre / Westfield / Intu / Goto or whatever we are calling Derby now!)
- You cannot mount Trunking across doorways!

Cable Containment – Cable Tray



Cable Containment - whats in the box??

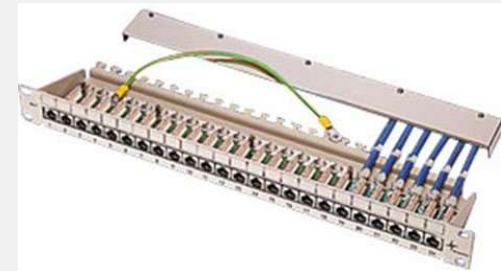
- We do not use a single piece of cable from an end device to a switch – if it became damaged we would need to replace it completely
- Instead, we build “extension cables” into the walls
- We add a cable to the trunking next to the user location, follow a path (horizontally then vertically) through the trunking, into the ceiling void, along a cable tray, up or down floors as required, down a cable tray and into the cabinet where the switches are

Cable Containment - whats in the box??

- Next we add an RJ45 (style) socket to each end of the cable, mount the user end in a socket box faceplate and the switch end into a 19inch patch panel, write the same number on each end!!!



Pin 1 – pin1, pin2 – pin2, ...



← Up to 100m →

Finally...

- Attach a flexible patch lead (RJ45 plug to plug) to the end user device, connect to the trunking mounted socket.
- Attach a flexible patch lead (RJ45 plug to plug) to the switch, connect to the patch panel socket
- End user device is now attached to switch
- If the end user device needs to be relocated, simply a case of relocating the patch leads (does not matter where in the building the device moves to!)

Extending the LAN

- If a wireless LAN is used to increase flexibility of connectivity, a WAP can be connected in a similar way to any other end user device
- Datapoint sockets are left available *above* the false ceiling, usually attached to the cable tray
- The WAP connects through the patch lead, building cable, patch lead scheme, PoE from the switch provides the power for the WAP

A word on Microsoft...

- The spec said MS servers and Clients and yet it hardly gets another mention...
- You need to make sure about the connection method but other than this there is little of relevance
- It is included to make it clear that you will be building a “standard” client server type network, you will not need any special servers or attachment methods

Do Not Copy This Example!

- This is similar but not the same as your design
- This example should help you to understand the design requirements, figure out what you need to do and identify some sources of information
- There are many types of twisted pair cables, many types of fibre and many wireless protocols
- You can use any network equipment but it must be able to satisfy the design, you must be able to explain why you are using it and it must be currently available (new, unused with full manufacturer warranty and support for at least two years)

Later weeks...

- Will be covering IP addressing in detail
- Will cover network layers
- Will cover communication protocols and processes

